

Dissertation (C.Sc. 666) Syllabus

Course Title: Dissertation

Course No: C.Sc. 666

Full Marks: 200

Pass Marks: 100

Nature of the Course: Dissertation

Credit Hours: 8

Course Description

Dissertation is a research paper of full mark 200 offered in the curriculum of the M.Sc. second year. A student, opting for the dissertation must prepare a proposal under the supervision of a supervisor and defend in the department. Once accepted, a student can start the work under the supervision of the supervisor. This work provides students with skill and knowledge in conducting research on fundamental and application aspects of computer science and information technology.

Introduction

All the students are required to successfully complete a research project as a part of their course. This is a major component of your degree, being worth 8 credits. The topic of the dissertation work must be relevant to M.Sc. CSIT degree. This is why it is important to agree in advance the topic of your dissertation with your supervisor and what it will entail. It is important to know that the degree is a science degree, and therefore all dissertations must be subject to scientific research.

Structure of the Report

The best dissertations and reports usually all follow much the same structure, as described here. The exact layout of dissertations tends to vary depending upon the nature of the material and the style of the author. It is recommended that you discuss this in detail with your supervisor. However the following might be considered to be a typical layout:

1. **Title page:** With a signed declaration that the dissertation is your own work.

2. **Abstract:** Giving a short (500 words max) overview of the work.
3. **Acknowledgements:** Thanking anyone who has helped you in any way.
4. **Table of contents:** Giving page numbers for all major section headings.
5. **Introduction:** Set the scenes, explain why you are doing this work and what is the problem being solved. Most importantly you should clearly explain what the aims and objectives of your work are.
6. **Related work:** Explain the current state of the art in your area. What works have other people done (published or commercial) that is relevant to yours.
7. **Methodology:** Explain what tools and technologies have you used. If you have collected data then explain how it is collected and analyzed.
8. **Description of the work:** Explain what exactly you have done. If this is a software project, describe your software in detail. If it is a data-based project, present and explain your data in detail.
9. **Discussion:** Explain what your work means. In a software project you should evaluate the functionality of your software. In a research project you should interpret your experimental results. In all cases you should evaluate what you have achieved against the aims and objectives you outlined in the introduction. The discussion should always end with a Conclusions section - in which you should briefly explain what conclusions you have come to as a result of doing this work.
10. **References:** provide a list of papers, books and other publications that are explicitly referred to in the text. This list must be prepared alphabetically by last name of the author and the list must be numbered sequentially. For example, "Bishop, M. and Boneh, D., Elements of Computer Security, Pearson Education., 2009".
11. **Appendices:** Supplementary material should be included in appendices - these are optional, but they might contain:
 - a. Code listings – A listing of the code you have written for the project. You should NOT include code listings for code you have not written!! If your project involves modifying code previously written by others, then you may include

this other code as long as you indicate clearly in the code listing what parts have been written by you.

b. User manuals

c. Technical documentation

d. Raw data – If your dissertation involved data collection then this should usually be included in appendices. This should provide supporting evidence for claims made in the main part of the dissertation (eg copies of a user evaluation questionnaire and some sample responses).

e. Examples of test data

f. Electronic material on a floppy disk or CD taped inside the back cover. This might contain executable software, source code, graphics, slides used for your presentation, etc. Where the appendices are long (e.g. code listings) do not print them out, rather provide them on a CD.

Citations

Whenever materials from the list of references are used in the report it must be followed by a citation which is simply the number of the reference in the list enclosed in square brackets. For example, a reference to the third article listed in the list of references would contain the citation [3]. Multiple citation numbers can be incorporated within one citation when required. For example, references to the fourth, eighth, and eleventh entries in the reference list would appear as [4, 8, 11].

Report Format

13. The paper can be prepared using a word processor or LATEX. The students are highly recommended to use LATEX.
14. Margins - All margins must be one inch.
15. The text must be spaced by 1.5.

16. The text must be typed in 12 point font. The text must be typed in Times New Roman font.